

Madhav Wadhvana

AI Engineer | Python Developer | React Native Developer
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PROFILE

Computer Engineering graduate specializing in Artificial Intelligence, with hands-on industry experience building AI-powered backend services, mobile applications, and data pipelines. At Softrefine Technology, developed LLM-powered workflows and Flask APIs for content-generation platforms. Earlier, contributed to geospatial visualization and BI dashboards at VisiLean as a Product Development Intern. Holds two filed patents in agri-tech and healthcare automation. Seeking a software engineering or AI engineer role where deep Python, React Native, and applied ML skills translate directly into impactful products.

CORE SKILLS

Languages: C/C++, Python, Java (Beginner), HTML, SQL

Libraries / Frameworks: PyTorch, TensorFlow, OpenCV, Flask, Django, Spring Boot (Beginner), Git, Docker

Frontend: React, React Native, JavaScript, TypeScript, Tailwind CSS, Redux

Backend & APIs: Flask, Flask, Django, REST APIs

Databases: PostgreSQL, MongoDB, DynamoDB

Generative AI: LLM Applications, Prompt Engineering, Structured Output Generation, Hugging Face, API-based LLM Integrations

Cloud & DevOps: AWS (EC2, SageMaker), Azure, Docker, CI/CD pipelines

Tools: Visual Studio, Microsoft 365, Power BI, GitHub

PROFESSIONAL EXPERIENCE

AI Engineer | Python Developer

Dec 2025 – Present

Softrefine Technology Pvt. Ltd.

Ahmedabad, Gujarat

- Developed AI-powered backend services and Flask APIs for content-generation platforms, enabling reliable LLM-driven output at scale.
- Integrated LLM-powered workflows and structured output validation pipelines to ensure consistency between AI responses, templates, and frontend rendering.
- Designed JSON schema validation layers to enforce compatibility across AI outputs, templates, and UI components.
- Built template-driven rendering systems using HTML/Jinja with dynamic asset injection for flexible content delivery.
- Debugged and optimized production AI-enabled backend services for scalability, performance, and long-term maintainability.

React Native Developer (Part-time)

Jul 2025 – Dec 2025

Timeverse Ltd.

Ahmedabad, Gujarat

- Developed and maintained the full application stack covering both React Native frontend and backend services.
- Built scalable mobile experiences with smooth navigation, animations, and robust state management.
- Designed and implemented reusable React components; optimized performance and ensured pixel-perfect UI across devices.
- Developed and integrated RESTful APIs, managed database operations, and set up deployment pipelines for production-ready systems.
- Collaborated with cross-functional teams to deliver end-to-end features from concept to deployment.

Product Development Intern

VisiLean Pvt. Ltd.

Jul 2024 – Oct 2024

Ahmedabad, Gujarat

- Leveraged GitHub GraphQL API for data ingestion and ETL processing, generating JSON-based analytical reports to enhance repository intelligence.
- Implemented a Cesium JS-powered 3D geospatial visualization module, augmenting UI/UX interactivity for geospatial analytics.
- Refactored and optimized Power BI dashboards, streamlining data pipelines and enhancing BI visualizations for data-driven decision-making.
- Collaborated with cross-functional Agile squads throughout the SDLC, driving seamless CI/CD workflows from design inception to production rollout.

SELECTED PROJECTS

LLM-Powered AI Assistant. Built a cross-platform React Native app integrated with DeepSeek API for real-time AI-generated responses with streaming. Implemented context-aware conversational features using React Native Reanimated, React Context API, and Tailwind CSS.

3D Globe Integration. Developed and integrated a 3D globe feature into a web application using Cesium JS, enabling interactive geographical visualizations that are fully responsive across devices.

Credit Card Input Component. Built and customized a React Native credit card input component with real-time validation and smooth animations to enhance secure mobile payment experiences.

Billing System (Java Swing & JDBC). Engineered a billing system enabling users to manage transactions, generate invoices, and maintain a sales database efficiently.

Attendance Tracker (JSP, JDBC, HTML & CSS). Developed an attendance tracker facilitating simplified tracking and management of records for educational or organizational use.

PATENTS

Plantation and Habitat Control Device for Vegetative Propagated Crops | Patent No: 2023210141007A | Filed Jul 16, 2025

- A cuboidal device with motorized wheels, storage chamber, screw conveyor, and motorized cutter to prepare crop segments for sowing.
- Features a soil quality assessment module for real-time ground condition analysis, plus fertilizer and water systems regulated through valves, and a robotic arm for automated sowing.

Mobile Device for Rejuvenating Diabetic Patients | Patent No: 202521068003 | Filed Jul 16, 2025

- A mobile, wheeled device capturing user profile, prescriptions, and health goals; uses image sensing for identification and a wearable band for real-time vitals tracking.
- Generates customized diets by comparing prescriptions, vitals, and goals against a database; includes blood glucose monitor, smart glasses for guided exercise, and robotic arms for physical support.

EDUCATION

Bachelor of Engineering, Computer Engineering (Specialization: Artificial Intelligence)

Marwadi University, India

Jul 2021 – May 2025

Relevant Coursework: Machine Learning, Deep Learning, Data Science, NLP, Representation Learning, Large Language Models.

LANGUAGES

English (Professional) | Gujarati (Native) | Hindi (Fluent)